

Nuclear-Norm Best Approximation at Rank-Deficient Matrix-Line Crossings

Certificate Geometry and Singular Smoothing

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Abstract

We study nuclear-norm best approximation from a matrix line and its dual spectral-norm ball intersected with one hyperplane. At a rank-deficient minimizer, exact certificates form a tangency slice of the nuclear-norm subdifferential. We classify its feasibility and uniqueness, derive its minimum-Frobenius member by singular-value water-filling, and characterize when clipping is unnecessary. An admissible Fenchel family unifies two centers: the Huber regularizer has the water-filled center, whereas pseudo-Huber generally selects a different smooth center. A polynomial derivative tail of order p yields boundary exponent $p/(p+1)$, while finite saturation yields a linear law. For arbitrary matrix lines, we derive the regular expansion and analytic multiplier and direction expansions at strict rank-deficient minimizers of arbitrary corank, with computable first corrections. Compact-polar stationarity produces cubic/quadratic superconvergence and an explicit quintic/quartic refinement after a second cancellation. At transverse square corank-one boundaries, pseudo-Huber obeys the exact two-thirds law, and a moving-crossing normal form gives a uniform cubic balance through the strict-to-boundary transition. Finally, the compact-polar selection fails on a nonempty open set even when the approximant and exact certificate are unique.

Keywords. Nuclear norm; best approximation; spectral norm; Birkhoff–James orthogonality; polar factor; Fenchel smoothing; singular perturbation.

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1 Introduction

Let $G, \Theta \in \mathbb{R}^{m \times n}$, with $\Theta \neq 0$. Our primary object is the nuclear-norm distance from G to the one-dimensional matrix subspace generated by Θ :

$$d^* = \min_{\lambda \in \mathbb{R}} \phi(\lambda), \quad \phi(\lambda) = \|G + \lambda\Theta\|_* = \text{dist}_{\|\cdot\|_*}(G, \text{span}\{\Theta\}). \quad (1)$$

Classical distance duality identifies the same value with

$$p^* = \max_{\Phi} \{ \langle G, \Phi \rangle : \|\Phi\|_2 \leq 1, \langle \Theta, \Phi \rangle = 0 \}. \quad (2)$$

Thus $d^* = p^*$: the dual unit ball is the spectral-norm ball, and the annihilator of $\text{span}\{\Theta\}$ is one hyperplane. We study what happens when the affine matrix line $\lambda \mapsto G + \lambda\Theta$ meets the rank-deficient locus. The distinctive difficulty is that the natural point derivative is expressed by the compact polar factor. It is smooth on fixed-rank strata, but under general rank-changing perturbations the zero-kernel completion need not have a continuous extension; see, e.g., the classical polar-decomposition framework of Higham [1986].

When $\Theta = uv^\top$, (2) is the tangent LMO on the smooth spectral sphere. Norm-ball LMOs underlie matrix-aware first-order methods [Pethick et al., 2025]. Xie et al. [2026] introduced the corresponding compact-polar bisection, and Li et al. [2026, Appendix P.5] interpreted it as the spectral-sphere instance of an intrinsic LMO. Neither work treats rank-deficient crossings.

Classical boundary and the new object. Optimality in (1) is classical Birkhoff–James orthogonality. Distance duality, matrix-norm subdifferentials, and the associated criteria are established in [Singer, 1970, Watson, 1992, Ziętak, 1993, Watson, 1993, Bhatia and Šemrl, 1999, Grover, 2014, Johnston et al., 2022, Grover and Gupta, 2026]; we claim no novelty for those facts. The new object is the tangency slice $\partial\|A^\star\|_* \cap \Theta^\perp$, its spectral centers, and the singular asymptotics of their regularization paths.

Canonical approximant selection by Schatten limits is also classical: Legg and Ward [1985] treat $p \downarrow 1$, while Ziętak [2017] surveys the $p \uparrow \infty$ strict-spectral counterpart, including its conjectural general form. Our selection instead occurs on the equality-constrained certificate face. If $A = U_r \Sigma_r V_r^\top$ is a compact singular value decomposition, the compact polar factor is

$$S_c(A) = U_r V_r^\top. \quad (3)$$

At full rectangular rank, ϕ is differentiable and $\phi'(\lambda) = \langle \Theta, S_c(G + \lambda\Theta) \rangle$. This suggests the ordinary scalar equation

$$h_c(\lambda) := \langle \Theta, S_c(G + \lambda\Theta) \rangle = 0. \quad (4)$$

At full rectangular rank this equation is valid. When the affine matrix loses rank, it need not remain valid, because the complete subdifferential is

$$\partial\|A\|_* = \{U_r V_r^\top + U_0 K V_0^\top : \|K\|_2 \leq 1\}, \quad (5)$$

where U_0, V_0 span the left and right null spaces. The compact factor is only the choice $K = 0$. The omitted kernel block can be exactly what is needed to enforce $\langle \Theta, \Phi \rangle = 0$. Consequently, h_c can be monotone and still jump across zero.

Thus the LMO and nuclear-norm duality remain exact; only the replacement of a set-valued optimality graph by one compact selection can fail.

1.1 Contributions and novelty boundary

The contributions, in the order used by the analysis, are as follows.

1. Starting from the classical interval $\partial\phi = [a - \beta, a + \beta]$ [Watson, 1992, Lewis, 1995], we classify feasibility and uniqueness of the tangency slice. Its minimum-Frobenius member follows by singular-value water-filling; clipping is universally unnecessary exactly for scalar multiples of partial isometries.
2. An admissible Fenchel family unifies the hard and smooth centers: Huber gives K_F , pseudo-Huber gives K_R , and we characterize their agreement. A polynomial derivative tail of order $p > 1$ yields exponent $p/(p+1)$, while finite saturation gives a linear law. The novelty beyond standard smoothing principles [Nesterov, 2005, Lobos et al., 2026, Attouch, 1984] is the constrained spectral center and its coupling to the singular multiplier.
3. We prove the regular expansion and, at every strict rank-deficient minimizer, analytic multiplier and returned-direction expansions without shape, normal-rank, or corank restrictions. Compact-root instances have computable cubic/quadratic superconvergence and a quintic/quartic refinement. At transverse square corank-one boundaries, pseudo-Huber has the exact two-thirds law and a uniform cubic strict-to-boundary crossover. Our new content is the matrix-line exponents, constants, and crossover, rather than the existence of fractional central-path rates [Basu and Mohammad-Nezhad, 2022].
4. Finally, a strict 2×2 witness extends to a nonempty open compact-polar failure set in every square dimension at least two, even with unique approximant and certificate. This motivates retaining the full slice. Rank-one normals also admit an SVD-free crossing formula.

An open failure set implies neither an event-frequency law nor an outer algorithmic or learning-performance gain; those require additional models.

1.2 Organization

Sections 2 and 3 give the scalar interval, structural failure, and certificate geometry; Sections 4 and 5 develop the Fenchel centers and local laws; Section 6 verifies the constants. Auxiliary level-set geometry and certificate repair are collected in the supplement.

2 Exact duality and scalar optimality

We use the Frobenius pairing $\langle X, Y \rangle = \text{tr}(X^\top Y)$. The spectral and nuclear norms are dual. For a compact SVD $A = U_r \Sigma_r V_r^\top$, let U_0, V_0 be orthonormal null-space bases; a zero-dimensional factor is allowed. The standard formula (5) will be used without a choice of bases inside repeated positive singular spaces.

Theorem 2.1 (Strong duality and the complete solution face). *For arbitrary $G \in \mathbb{R}^{m \times n}$ and nonzero $\Theta \in \mathbb{R}^{m \times n}$,*

$$p^* = \max_{\substack{\|\Phi\|_2 \leq 1 \\ \langle \Theta, \Phi \rangle = 0}} \langle G, \Phi \rangle = \min_{\lambda \in \mathbb{R}} \|G + \lambda \Theta\|_*. \quad (6)$$

The dual minimizer set is nonempty and compact. If λ^ is any dual minimizer and $A^* = G + \lambda^* \Theta$, then all primal solutions are*

$$\mathcal{P}^* = \partial \|A^*\|_* \cap \Theta^\perp. \quad (7)$$

Every dual minimizer satisfies

$$|\lambda^*| \leq \frac{2\|G\|_*}{\|\Theta\|_*}. \quad (8)$$

Proof. The support function of the spectral unit ball is the nuclear norm. Since $\Phi = 0$ is feasible and lies in the interior of the ball, finite-dimensional Slater–Fenchel duality gives (6). The reverse triangle inequality $\|G + \lambda \Theta\|_* \geq |\lambda| \|\Theta\|_* - \|G\|_*$ gives coercivity and, by comparison with $\lambda = 0$, (8). Finally, equality in the support function is equivalent to $\Phi \in \partial \|A^*\|_*$; imposing tangency yields exactly (7). $\square$

At a fixed λ , set

$$A = G + \lambda \Theta = U_r \Sigma_r V_r^\top, \quad a = \langle \Theta, U_r V_r^\top \rangle, \quad C = U_0^\top \Theta V_0, \quad \beta = \|C\|_*. \quad (9)$$

Theorem 2.2 (Exact scalar interval and compact-root criterion). *For $\phi(\lambda) = \|G + \lambda \Theta\|_*$,*

$$\partial \phi(\lambda) = [a - \beta, a + \beta]. \quad (10)$$

Consequently, at this specified λ :

1. λ is dual optimal if and only if $|a| \leq \beta$;
2. the compact equation (4) holds if and only if $a = 0$;
3. dual optimality holds but compact stationarity fails if and only if $0 < |a| \leq \beta$.

Globally, (4) has no solution if and only if $a(\lambda) \neq 0$ for every $\lambda \in \arg \min \phi$. If this happens, the dual minimizer is unique and its affine matrix is rank deficient.

Proof. The affine chain rule and (5) give $\partial \phi(\lambda) = \{a + \langle C, K \rangle : \|K\|_2 \leq 1\}$, whose endpoints are $a \pm \|C\|_*$. This proves (10) and the three pointwise claims. If two minimizers existed, convexity would make ϕ constant between them; at every interior point $\partial \phi = \{0\}$, hence $a = \beta = 0$ and a compact root would exist. Thus no-root failure forces uniqueness. Full rectangular rank would give $\beta = 0$, and optimality would again force $a = 0$, so the unique no-root minimizer is rank deficient. $\square$

Remark 2.3 (Relative-interior nondegeneracy). At an optimum, $|a| < \beta$ is exactly $0 \in \text{ri } \partial \phi(\lambda^*)$; equivalently, the tangency hyperplane meets the relative interior of the kernel-block spectral unit ball. At $|a| = \beta$ it meets only the exposed endpoint face (21). This parallels convergence and limiting-center phenomena for degenerate semidefinite central paths [Halická et al., 2002, 2005], while fractional algebraic rates are documented by Basu and Mohammad-Nezhad [2022]. Our rates follow directly from the matrix-line normal forms in Section 5.

The compact selection is nevertheless nondecreasing. Indeed, for $\lambda_1 < \lambda_2$, each compact polar factor belongs to the corresponding nuclear-norm subdifferential, and monotonicity of the convex subdifferential graph gives $h_c(\lambda_1) \leq h_c(\lambda_2)$. The obstruction is a jump, not a reversal of monotonicity.

Remark 2.4 (Exact interval bisection). With $L = a - \beta$ and $U = a + \beta$, $U < 0$ moves the bracket to the right, $L > 0$ moves it to the left, and $L \leq 0 \leq U$ certifies a minimizer. The initial bracket is (8); on termination one applies the completion (22). Unlike compact-polar bisection, this test cannot jump over a minimizer, but it requires a numerical-rank decision. The elementary convergence proof is recorded in the supplement.

3 Structural failure and certificate geometry

The rank-one normal from the motivating tangent oracle supplies a transparent witness. The structural result below then perturbs that witness in the full space of matrix normals; the completion theory is rectangular and makes no rank assumption on Θ .

3.1 A witness and full-space openness

Let $b_1, b_2, c > 0$, $d \in \mathbb{R}$, and

$$\Theta = e_1 e_1^\top, \quad G = \begin{pmatrix} d & b_1 \\ b_2 & c \end{pmatrix}. \quad (11)$$

Writing $x = d + \lambda$, set $A(x) = \begin{pmatrix} x & b_1 \\ b_2 & c \end{pmatrix}$.

Theorem 3.1 (A strict 2×2 family with no compact root). *If $0 < b_1 b_2 \leq c^2$, then*

$$x^\star = \frac{b_1 b_2}{c} \quad (12)$$

is the unique minimizer of $\|A(x)\|_$, and the compact scalar selection does not attain zero. At the singular minimizer,*

$$\langle e_1 e_1^\top, S_c(A(x^\star)) \rangle = \frac{b_1 b_2}{\sqrt{(b_1^2 + c^2)(b_2^2 + c^2)}} > 0. \quad (13)$$

The strict condition $0 < b_1 b_2 < c^2$ is open in the positive parameter orthant.

Proof. For a 2×2 matrix, $\|A\|_*^2 = \|A\|_F^2 + 2|\det A|$. Since $\det A(x) = xc - b_1 b_2$,

$$\|A(x)\|_*^2 = \begin{cases} (x - c)^2 + (b_1 + b_2)^2, & x < x^\star, \\ (x + c)^2 + (b_1 - b_2)^2, & x > x^\star. \end{cases} \quad (14)$$

Below the crossing, the derivative of the nuclear norm is $(x - c)/\sqrt{(x - c)^2 + (b_1 + b_2)^2}$. Above it, the derivative is $(x + c)/\sqrt{(x + c)^2 + (b_1 - b_2)^2}$. They are respectively negative and positive because $x^\star \leq c$, proving uniqueness. Moreover

$$A(x^\star) = \begin{pmatrix} b_1 \\ c \end{pmatrix} \begin{pmatrix} b_2/c & 1 \end{pmatrix},$$

whose compact polar has the entry in (13). The compact scalar therefore jumps from a negative to a positive value without attaining zero. $\square$

The equality case $b_1 b_2 = c^2$ will be the boundary-kink model in Section 5. Strict instances give a stronger statement than relative openness inside the rank-one manifold. Write

$$\mathbb{S}_F^{n^2-1} = \{\Theta \in \mathbb{R}^{n \times n} : \|\Theta\|_F = 1\}.$$

Theorem 3.2 (Full-space openness of strict compact-polar failure). *For every $n \geq 2$, there are a nonempty relatively open set*

$$\mathcal{O}_n \subset \text{GL}_n \times \mathbb{S}_F^{n^2-1}$$

and a real-analytic map $(G, \Theta) \mapsto \lambda_s(G, \Theta)$ such that $A_s = G + \lambda_s \Theta$ has rank $n - 1$. With unit left and right null vectors u_s, v_s , define

$$a_s = \langle \Theta, S_c(A_s) \rangle, \quad \beta_s = |u_s^\top \Theta v_s|. \quad (15)$$

Then

$$0 < |a_s| < \beta_s. \quad (16)$$

The crossing λ_s is the unique global minimizer and satisfies the sharp-growth bound

$$\|G + \lambda \Theta\|_* \geq \|A_s\|_* + (\beta_s - |a_s|)|\lambda - \lambda_s| \quad (\lambda \in \mathbb{R}). \quad (17)$$

The compact equation (4) has no solution. In particular, strict failure has nonempty open interior in the full normalized data space, which includes full-rank matrix normals.

Proof. For $n = 2$, take

$$G_0 = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \quad \Theta_0 = e_1 e_1^\top, \quad \lambda_0 = \frac{1}{2}.$$

Then G_0 is invertible and $A_0 = \frac{1}{2}(1, 2)^\top(1, 2)$. Its normalized range and null vectors are $w = (1, 2)^\top/\sqrt{5}$ and $z = (-2, 1)^\top/\sqrt{5}$, respectively, so

$$a_0 = \langle \Theta_0, ww^\top \rangle = \frac{1}{5}, \quad \beta_0 = |z^\top \Theta_0 z| = \frac{4}{5}.$$

For $n > 2$, embed G_0 as the leading block of $\text{diag}(G_0, d_3, \dots, d_n)$, with $d_j > 0$, and extend Θ_0 by zeros. The crossing has corank one and retains these strict values.

Set $F(G, \Theta, \lambda) = \det(G + \lambda\Theta)$. At the witness, $\partial_\lambda F \neq 0$, because at a corank-one matrix the adjugate is a nonzero multiple of $v_s u_s^\top$ and $u_s^\top \Theta v_s \neq 0$. The analytic implicit-function theorem, restricted to the Frobenius sphere, therefore continues the crossing as $\lambda_s(G, \Theta)$. After shrinking the neighborhood, $\partial_\lambda F \neq 0$. Jacobi's formula gives

$$\partial_\lambda \det A = \text{tr}(\text{adj}(A)\Theta),$$

whereas $\text{adj}(A) = 0$ whenever $\text{rank } A \leq n - 2$. Thus a zero of F with $\partial_\lambda F \neq 0$ has rank exactly $n - 1$, so the continued crossing remains corank one. The partial polar factor and the null projectors are continuous along this fixed-rank crossing manifold, so a_s and β_s are continuous and the strict inequalities persist.

At the crossing, [theorem 2.2](#) gives $\partial\phi(\lambda_s) = [a_s - \beta_s, a_s + \beta_s]$. Apply the subgradient inequality with the right endpoint for $\lambda > \lambda_s$ and the left endpoint for $\lambda < \lambda_s$; both imply (17). Thus the minimizer is globally unique. A compact-polar root anywhere would provide a zero scalar subgradient and hence another minimizer; at λ_s the compact scalar equals $a_s \neq 0$. No root exists. Finally, full-rank normals are dense in $\mathbb{S}_F^{n^2-1}$, so $\mathcal{O}_n$ contains such normals. $\square$

The implicit-function theorem asserts local uniqueness of the continued *crossing*, not global uniqueness of all determinant roots; small general-normal perturbations may create additional far-away crossings. Equation (17) is what makes the continued crossing the unique global *minimizer*. The theorem is topological and makes no frequency claim.

For the rank-one normal appearing in the tangent-oracle application, the crossing and its half-width admit an SVD-free formula. Let

$$\Theta = uv^\top, \quad \|u\|_2 = \|v\|_2 = 1. \quad (18)$$

Proposition 3.3 (Rank-one crossing formula). *If $G \in \text{GL}_n$ and $\alpha = v^\top G^{-1}u \neq 0$, the rank-one line has a corank-one crossing at $\lambda_{\text{sing}} = -\alpha^{-1}$, and*

$$\beta = \frac{\alpha^2}{\|G^{-1}u\|_2 \|G^{-\top}v\|_2}. \quad (19)$$

Proof. The determinant lemma gives $\det(G + \lambda uv^\top) = \det(G)(1 + \lambda\alpha)$. The right and left null spaces at its root are spanned by $G^{-1}u$ and $G^{-\top}v$. After normalization, both numerators in $\beta = |u_0^\top u| |v_0^\top v|$ equal $|\alpha|$, which proves (19). The factor $I - \alpha^{-1}G^{-1}uv^\top$ is the identity minus a rank-one projector, so the corank is one. $\square$

3.2 The complete certificate slice and its uniqueness

Fix a dual minimizer and write

$$A^* = G + \lambda^* \Theta = U_r \Sigma_r V_r^\top, \quad \ell = m - r, \quad k = n - r, \quad C = U_0^\top \Theta V_0, \quad \beta = \|C\|_*.$$

By (7), every exact certificate is

$$\Phi^* = U_r V_r^\top + U_0 K V_0^\top, \quad K \in \mathcal{K}(C, a) := \{K : \|K\|_2 \leq 1, \langle C, K \rangle = -a\}. \quad (20)$$

Theorem 3.4 (Feasibility and uniqueness of the certificate slice). *The slice $\mathcal{K}(C, a)$ is nonempty if and only if $|a| \leq \beta$. If $\ell k = 0$, then $C = 0$, the kernel-block space consists only of the empty zero matrix, and the slice is feasible and a singleton exactly when $a = 0$. Assume henceforth that $\ell, k \geq 1$. If $C = 0$, feasibility forces $a = 0$, and the slice is the entire, non-singleton spectral unit ball. If $C \neq 0$, then:*

1. if $|a| < \beta$, the slice is a singleton if and only if $\ell k = 1$;
2. if $|a| = \beta$, the slice is a singleton if and only if $\text{rank } C = \min\{\ell, k\}$.

In all remaining feasible cases with $\ell, k \geq 1$, it contains infinitely many matrices. Consequently, at a transverse square corank-one crossing the exact certificate is unique in both the strict and boundary regimes.

Proof. Feasibility is spectral–nuclear duality. The zero-dimensional case is immediate. If $\ell, k \geq 1$ and $C = 0$, the slice is the whole spectral unit ball. Now let $r_C = \text{rank } C$, let $C = P_C \Sigma_C Q_C^\top \neq 0$ be a compact SVD, and put $S_C = P_C Q_C^\top$. When $|a| < \beta$, the feasible matrix $K_0 = -(a/\beta)S_C$ has spectral norm strictly below one. If $\ell k > 1$, choose nonzero $H \perp C$; both $K_0 \pm \varepsilon H$ remain feasible for small ε . If $\ell k = 1$, the equality fixes the only entry.

At $|a| = \beta$, equality in spectral–nuclear duality gives the complete endpoint face

$$\mathcal{K}(C, a) = \{-\text{sign}(a)(P_C Q_C^\top + P_\perp W Q_\perp^\top) : \|W\|_2 \leq 1\}, \quad (21)$$

where $W \in \mathbb{R}^{(\ell-r_C) \times (k-r_C)}$, and $P_\perp, Q_\perp$ complete the singular bases. Indeed, equality for each positive singular pair forces $K q_i = -\text{sign}(a) p_i$ and $K^\top p_i = -\text{sign}(a) q_i$, which kills both cross blocks. The face is a singleton exactly when one complementary space is zero, equivalently when $\text{rank } C = \min\{\ell, k\}$. $\square$

Thus compact-polar failure is not an artifact of certificate nonuniqueness: in the square corank-one instances governing Section 5, both the best approximant and the exact certificate are unique, but the compact choice $K = 0$ is still wrong whenever $a \neq 0$.

3.3 Hard spectral completion and the no-clipping dichotomy

The minimum-Frobenius member of the complete slice can always be computed from one SVD and one monotone scalar equation. This also supplies step 4 of the exact interval oracle.

Theorem 3.5 (Minimum-Frobenius spectral completion). *Let $C \in \mathbb{R}^{\ell \times k}$ and suppose $|a| \leq \beta := \|C\|_*$. If $C = 0$, then $a = 0$ and the unique minimum-Frobenius completion is $K_F = 0$. Otherwise, let $C = P_C \text{diag}(\sigma_1, \dots, \sigma_{r_C}) Q_C^\top$ be a compact SVD, where $r_C = \text{rank } C$ and $\sigma_1 \geq \dots \geq \sigma_{r_C} > 0$. Define*

$$g_F(\eta) = \sum_{i=1}^{r_C} \sigma_i \min\{\eta \sigma_i, 1\}.$$

For $|a| < \beta$, there is a unique $\eta \in [0, 1/\sigma_{r_C})$ with $g_F(\eta) = |a|$; at the endpoint take $\eta = 1/\sigma_{r_C}$. With $\text{sign}(0) = 0$, the unique member of $\mathcal{K}(C, a)$ of minimum Frobenius norm is

$$K_F = -\text{sign}(a) P_C \text{diag}(\min\{\eta \sigma_i, 1\}) Q_C^\top. \quad (22)$$

At $|a| = \beta$, the multiplier parameter need not be unique, but the displayed matrix is unique.

Proof. The case $C = 0$ is immediate. Otherwise, projection of μC onto the spectral unit ball clips its singular values at one. The continuous function g_F increases strictly until it reaches β , so the stated multiplier exists. With $\mu = -\text{sign}(a)\eta$, this projection is K_F and satisfies the equality. For any other feasible L , projection optimality and $\langle C, L - K_F \rangle = 0$ give $\langle K_F, L - K_F \rangle \geq 0$. Hence

$$\|L\|_F^2 \geq \|K_F\|_F^2 + \|L - K_F\|_F^2,$$

which proves optimality and uniqueness. $\square$

For the rank-one normal (18), put $x = U_0^\top u$, $y = V_0^\top v$. Then $C = xy^\top$ and $\beta = \|x\|_2 \|y\|_2$. If $\beta > 0$, no clipping occurs, and

$$K_F = K_{\text{can}} = -\frac{a}{\beta} \frac{x}{\|x\|_2} \frac{y^\top}{\|y\|_2}. \quad (23)$$

If $\beta = 0$, dual optimality forces $a = 0$, and $K_F = K_{\text{can}} = 0$.

Proposition 3.6 (No-clipping/partial-isometry dichotomy). *For fixed $C \neq 0$, the unconstrained minimum-Frobenius solution*

$$K_{\text{lin}} = -\frac{a}{\|C\|_F^2} C$$

is spectrally feasible if and only if $|a| \leq \|C\|_F^2 / \|C\|_2$. Moreover

$$\frac{\|C\|_F^2}{\|C\|_2} \leq \|C\|_*, \quad (24)$$

with equality if and only if all nonzero singular values of C coincide. Equivalently, K_{lin} is feasible for every admissible scalar right-hand side if and only if C is a scalar multiple of a partial isometry.

Proof. The first claim follows from $\|K_{\text{lin}}\|_2 = |a| \|C\|_2 / \|C\|_F^2$. If $\sigma_1 = \|C\|_2$, then

$$\|C\|_F^2 = \sum_i \sigma_i^2 \leq \sigma_1 \sum_i \sigma_i = \|C\|_2 \|C\|_*,$$

with equality exactly when each positive σ_i equals σ_1 . $\square$

For a general normal, (22) replaces the rank-one formula by one SVD and one scalar solve. The smooth analogue below has a different spectral profile; corollary 4.4 gives the exact condition under which the two centers agree.

4 Fenchel recovery and center selection

For $\delta > 0$, define

$$\psi_\delta(s) = \sqrt{s^2 + \delta^2} - \delta, \quad \Psi_\delta(A) = \sum_{i=1}^q \psi_\delta(\sigma_i(A)), \quad q = \min\{m, n\}. \quad (25)$$

This is a pseudo-Huber smoothing of the nuclear norm. The spectral Fenchel pair, its gradient formula, and its Frobenius Lipschitz continuity are known [Mustafi et al., 2026, Lobos et al., 2026, Feoktistov et al., 2026]; we use that pair after imposing the tangency equality to identify the constrained regularized limit inside (20).

The scalar conjugate is

$$\psi_\delta^*(z) = \begin{cases} \delta(1 - \sqrt{1 - z^2}), & |z| \leq 1, \\ +\infty, & |z| > 1. \end{cases} \quad (26)$$

Indeed, for $|z| < 1$, stationarity in the conjugate gives $s = \delta z / \sqrt{1 - z^2}$, and the endpoints follow by continuity. Define

$$R(\Phi) = \sum_{i=1}^q \left(1 - \sqrt{1 - \sigma_i(\Phi)^2}\right) + \iota_{\{\|\Phi\|_2 \leq 1\}}(\Phi). \quad (27)$$

Lemma 4.1 (Conjugacy, strong convexity, and smooth gradient). *The regularizer R is 1-strongly convex in the Frobenius norm, and*

$$\Psi_\delta = (\delta R)^*. \quad (28)$$

The function Ψ_δ is C^∞ and strictly convex on the full matrix space. If $A = U \text{diag}(\sigma_i) V^\top$ is any full SVD, then

$$Z_\delta(A) := \nabla \Psi_\delta(A) = U \text{diag} \left(\frac{\sigma_i}{\sqrt{\sigma_i^2 + \delta^2}} \right) V^\top. \quad (29)$$

Equivalently,

$$Z_\delta(A) = \begin{cases} A(A^\top A + \delta^2 I)^{-1/2}, & m \geq n, \\ (AA^\top + \delta^2 I)^{-1/2} A, & m < n. \end{cases} \quad (30)$$

Proof. On $(-1, 1)$, the even scalar function $\rho(t) = 1 - \sqrt{1 - t^2}$ satisfies $\rho''(t) = (1 - t^2)^{-3/2} \geq 1$. Hence $\rho(t) - t^2/2$, extended by $+\infty$ outside $[-1, 1]$, is proper, closed, and convex. Spectral lifting of absolutely symmetric closed convex functions [Lewis, 1995] shows that $R(\Phi) - \|\Phi\|_F^2/2$ is proper, closed, and convex. Thus R is lower semicontinuous, and this is the asserted strong convexity. The same lifting applied to (26) proves (28).

For strict matrix convexity, use the symmetric dilation

$$\mathcal{H}(A) = \begin{pmatrix} 0 & A \\ A^\top & 0 \end{pmatrix}, \quad g_\delta(t) = \sqrt{t^2 + \delta^2}.$$

Its spectrum consists of $\pm\sigma_i(A)$ and $|m - n|$ zeros, so

$$\Psi_\delta(A) = \frac{1}{2} \operatorname{tr} g_\delta(\mathcal{H}(A)) - \frac{m + n}{2} \delta.$$

For a symmetric matrix, the trace-spectral Hessian has divided differences $(g'_\delta(x) - g'_\delta(y))/(x - y)$, with the diagonal value $g''_\delta(x)$ [Lewis and Sennov, 2001]. Since $g'_\delta(t) = t/\sqrt{t^2 + \delta^2}$ is strictly increasing, all these coefficients are positive. A nonzero matrix perturbation has a nonzero dilation, so the quadratic Hessian form is positive. This proves strict convexity; functional calculus gives smoothness and (29)–(30). $\square$

Theorem 4.2 (Unique smooth oracle and Fenchel center path). *For every $\Theta \neq 0$, the scalar function*

$$d_\delta(\lambda) = \Psi_\delta(G + \lambda\Theta) \tag{31}$$

is coercive and strictly convex. It has a unique minimizer λ_δ , characterized by

$$h_\delta(\lambda) := \langle \Theta, Z_\delta(G + \lambda\Theta) \rangle = 0. \tag{32}$$

The direction

$$\Phi_\delta = Z_\delta(G + \lambda_\delta\Theta) \tag{33}$$

is exactly tangent and satisfies $\|\Phi_\delta\|_2 < 1$.

Moreover, Φ_δ is the unique solution of

$$\max_{\substack{\|\Phi\|_2 \leq 1 \\ \langle \Theta, \Phi \rangle = 0}} \{ \langle G, \Phi \rangle - \delta R(\Phi) \}. \tag{34}$$

If $\mathcal{P}^$ is the exact solution face in (7), then*

$$\Phi_\delta \longrightarrow \Phi_R^* := \arg \min_{\Phi \in \mathcal{P}^*} R(\Phi) \quad (\delta \downarrow 0). \tag{35}$$

The center is unique. If the exact dual minimizer is unique, then $\lambda_\delta \rightarrow \lambda^$.*

Proof. Strict convexity follows by restricting Ψ_δ to the injective affine line $G + \lambda\Theta$. Since

$$0 \leq \|A\|_* - \Psi_\delta(A) \leq q\delta, \tag{36}$$

the lower bound $d_\delta(\lambda) \geq |\lambda| \|\Theta\|_* - \|G\|_* - q\delta$ proves coercivity. Differentiation gives (32); the derivative of a differentiable, strictly convex, coercive scalar function has one zero. Each multiplier in (29) lies in $[0, 1)$, and the root gives tangency, proving the first part.

Conjugacy (28) gives

$$\Psi_\delta(G + \lambda\Theta) = \max_{\|\Phi\|_2 \leq 1} \{ \langle G + \lambda\Theta, \Phi \rangle - \delta R(\Phi) \}.$$

The feasible set of (34) is compact and contains the relative Slater point zero. Lagrange duality is therefore exact, while strong convexity of R makes the primal optimizer unique. Fenchel equality identifies it with (33).

Let p^* denote the unregularized optimum. Optimality against the unique R -minimizer Φ_R^* on $\mathcal{P}^*$ gives

$$0 \leq p^* - \langle G, \Phi_\delta \rangle \leq \delta \{ R(\Phi_R^*) - R(\Phi_\delta) \}. \tag{37}$$

In particular, $R(\Phi_\delta) \leq R(\Phi_R^*)$ and, because $R \geq 0$, the linear gap is at most $\delta R(\Phi_R^*) \rightarrow 0$. Compactness makes every sequence $\delta_k \downarrow 0$ have a convergent subsequence. The linear gap in (37) puts every limit in $\mathcal{P}^*$; lower semicontinuity gives an R -value no larger than that of Φ_R^* . Strong convexity makes the center unique, so all subsequences have the same limit.

Finally, the uniform convergence (36) and common coercive lower bound imply convergence of the scalar minimizers whenever the exact minimizer is unique. $\square$

The abstract center has an explicit spectral form for every matrix normal. It is generally different from the hard minimum-Frobenius completion (22).

Theorem 4.3 (General pseudo-Huber certificate center). *At a rank-deficient optimum, use C , a , and β from (9). If $C = 0$, then $a = 0$ and $K_R = 0$. Otherwise, let $C = P_C \text{diag}(\sigma_1, \dots, \sigma_{r_C}) Q_C^\top$ be a compact SVD, where $r_C = \text{rank } C$. If $|a| < \beta$, there is a unique $\eta \geq 0$ satisfying*

$$\sum_{i=1}^{r_C} \frac{\eta \sigma_i^2}{\sqrt{1 + \eta^2 \sigma_i^2}} = |a|. \quad (38)$$

The kernel block of Φ_R^* is

$$K_R = -\text{sign}(a) P_C \text{diag} \left(\frac{\eta \sigma_i}{\sqrt{1 + \eta^2 \sigma_i^2}} \right) Q_C^\top, \quad (39)$$

with $K_R = 0$ when $a = 0$. At $|a| = \beta$, the formula is understood as $\eta \rightarrow \infty$, giving $K_R = -\text{sign}(a) P_C Q_C^\top$. For rank-one C , (39) reduces exactly to K_{can} in (23).

Proof. In the singular bases, $R(\Phi) = \text{rank}(A^*) + \mathcal{R}_0(K)$, where

$$\mathcal{R}_0(K) = \sum_j \rho(\sigma_j(K)) + \iota_{\{\|K\|_2 \leq 1\}}(K), \quad \rho(t) = 1 - \sqrt{1 - t^2}.$$

The scalar conjugate is $\rho^*(z) = \sqrt{1 + z^2} - 1$. Spectral lifting gives

$$\mathcal{R}_0^*(Y) = \sum_{j=1}^{\min\{\ell, k\}} (\sqrt{1 + \sigma_j(Y)^2} - 1).$$

For $|a| < \beta$, the feasible point $-(a/\beta) P_C Q_C^\top$ has spectral norm below one, so Slater holds. The KKT condition is $K_R = \nabla \mathcal{R}_0^*(\mu C)$, with $\mu = -\text{sign}(a)\eta$. It gives (39) and

$$\langle C, K_R \rangle = -\text{sign}(a) \sum_{i=1}^{r_C} \frac{\eta \sigma_i^2}{\sqrt{1 + \eta^2 \sigma_i^2}} = -a.$$

The left side of (38) starts at zero, has derivative

$$\sum_i \frac{\sigma_i^2}{(1 + \eta^2 \sigma_i^2)^{3/2}} > 0,$$

and tends to $\sum_i \sigma_i = \beta$, proving existence and uniqueness for the strict slice. At the endpoint, (21) fixes the support block and leaves only W ; nonnegativity of ρ , with equality only at zero, selects $W = 0$. The case $C = 0$ is the same unconstrained minimum. For rank-one C , the spectral center formula has one active coefficient, and (38) fixes it at $|a|/\beta$, yielding (23). $\square$

Corollary 4.4 (Exact agreement of the hard and smooth centers). *Let $C \neq 0$ and $|a| \leq \beta$. Then $K_F = K_R$ if and only if at least one of the following holds: $a = 0$, $|a| = \beta$, or all positive singular values of C coincide. Thus, in the strict nonzero regime, the two centers agree exactly for scalar multiples of partial isometries.*

Proof. Agreement is immediate at zero and at the endpoint. If all positive singular values are equal, symmetry makes every active coefficient in both formulas equal to $|a|/\beta$. Conversely, suppose $0 < |a| < \beta$ and the centers agree. Every coefficient in the smooth formula is strictly below one, so the hard formula is unsaturated. Equality of the coefficients gives, for every positive σ_i ,

$$\eta_F = \frac{\eta_R}{\sqrt{1 + \eta_R^2 \sigma_i^2}}.$$

Since $\eta_R > 0$, all σ_i must coincide. $\square$

4.1 A smoothing family and two distinguished centers

The preceding center is not an isolated artifact of pseudo-Huber smoothing. Call a scalar profile ψ *admissible* if it is even, convex, and C^1 , with $\psi(0) = 0$, if $0 \leq \psi'(x) \leq 1$ for $x \geq 0$, and if ψ' is nondecreasing with $\lim_{x \rightarrow \infty} \psi'(x) = 1$. We further require

$$c_\psi := \lim_{x \rightarrow \infty} \{x - \psi(x)\} < \infty, \quad 0 \leq x < y, \psi'(y) < 1 \implies \psi'(x) < \psi'(y), \quad (\text{A})$$

and assume that, for some $\mu_\psi > 0$, the extended-value function $z \mapsto \psi^*(z) - \mu_\psi z^2/2$ is convex on $\mathbb{R}$; the domain of ψ^* is $[-1, 1]$. The second condition in (A) means strict increase before possible finite saturation; it includes both profiles below. Define

$$\Psi_\delta^\psi(A) = \delta \sum_{i=1}^q \psi\left(\frac{\sigma_i(A)}{\delta}\right), \quad R_\psi(\Phi) = \sum_{i=1}^q \psi^*(\sigma_i(\Phi)). \quad (40)$$

The last expression is understood as $+\infty$ when $\|\Phi\|_2 > 1$.

Proposition 4.5 (Admissible profiles select spectral centers). *For every admissible ψ , $\Psi_\delta^\psi = (\delta R_\psi)^*$, and R_ψ is strongly convex in the Frobenius norm. If λ_δ^ψ is any minimizer of $\Psi_\delta^\psi(G + \lambda\Theta)$, then*

$$\Phi_\delta^\psi = \nabla \Psi_\delta^\psi(G + \lambda_\delta^\psi \Theta)$$

is independent of the choice of multiplier and is the unique maximizer of (34) with R replaced by R_ψ . Moreover,

$$\Phi_\delta^\psi \longrightarrow \Phi_\psi^* := \arg \min_{\Phi \in \mathcal{P}^*} R_\psi(\Phi). \quad (41)$$

At a strict rank-deficient minimizer, with a, C, β as in (9), every choice of λ_δ^ψ satisfies

$$\frac{\lambda_\delta^\psi - \lambda^*}{\delta} \longrightarrow t_0^\psi = -\text{sign}(a)\eta_\psi, \quad \sum_{i=1}^{r_C} \sigma_i \psi'(\eta_\psi \sigma_i) = |a|, \quad (42)$$

where $\eta_\psi \geq 0$ is the unique solution. The null-null block of the center is

$$K_\psi = -\text{sign}(a)P_C \text{diag}(\psi'(\eta_\psi \sigma_i))Q_C^\top. \quad (43)$$

The same sharp-growth argument gives the uniform localization

$$|\lambda_\delta^\psi - \lambda^*| \leq \frac{qc_\psi}{\beta - |a|} \delta. \quad (44)$$

For

$$\psi_{\text{ph}}(x) = \sqrt{1+x^2} - 1, \quad \psi_{\text{hub}}(x) = \begin{cases} x^2/2, & |x| \leq 1, \\ |x| - 1/2, & |x| > 1, \end{cases} \quad (45)$$

these formulas give, respectively, (η_R, K_R) and (η_F, K_F) . In particular,

$$R_{\text{hub}}(\Phi) = \frac{1}{2} \|\Phi\|_F^2 + \iota_{\{\|\Phi\|_2 \leq 1\}}(\Phi), \quad (46)$$

so the hard water-filled completion is itself a Fenchel center.

Proof. Scalar perspective conjugacy and spectral lifting give $\Psi_\delta^\psi = (\delta R_\psi)^*$; strong convexity also lifts from the absolutely symmetric singular-value function. Hence the proof of theorem 4.2, without its strict-convexity claim for the scalar multiplier, gives the unique regularized certificate and (41). The uniform bound $0 \leq |x| - \delta\psi(x/\delta) \leq c_\psi\delta$ replaces (36).

For the local rate, use the exact analytic cluster frames from theorem 5.4. After setting $s = \delta t$, the separated positive cluster contributes $at + o(1)$, whereas the small block contributes $\sum_i \psi(t\sigma_i) + o(1)$, uniformly for bounded t . Thus the rescaled objectives converge compact-uniformly to

$$F_\psi(t) = at + \sum_{i=1}^{r_C} \psi(t\sigma_i).$$

It is coercive because $|a| < \beta$. The map $g_\psi(t) = F'_\psi(t) - a$ is strictly increasing wherever $|g_\psi(t)| < \beta$ by (A). Since $|a| < \beta$, F'_ψ has the unique zero in (42). The sharp-growth localization used in theorem 5.2 then gives convergence of every minimizing multiplier. Finally, KKT conditions for minimizing the null-block part of R_ψ over $\mathcal{K}(C, a)$, or equivalently differentiating its spectral conjugate at $-\text{sign}(a)\eta_\psi C$, give (43). For pseudo-Huber, $\psi'_{\text{ph}}(x) = x/\sqrt{1+x^2}$, recovering (38)–(39). For Huber, $\psi'_{\text{hub}}(x) = \min\{x, 1\}$ on $[0, \infty)$ and $\psi^*_{\text{hub}}(z) = z^2/2$ on $[-1, 1]$, which gives (46); its scalar equation is exactly the hard water-filling equation defining η_F and K_F . $\square$

Remark 4.6 (Variational center versus analytic oracle). Huber smoothing is C^1 , convex, and coercive, so its scalar stationarity equation always has a solution. Its derivative saturates, however, so the objective need not be strictly convex and can have locally constant multiplier intervals; the clipping map is also nonanalytic at its threshold. Thus multiplier uniqueness and the analytic direction expansion of theorem 5.4 do not transfer without extra active-set nondegeneracy. Huber is used here to identify K_F variationally and to separate center selection from the tail mechanism governing boundary rates, not to replace the pseudo-Huber oracle. For example, if $G = 2I_2$, $\Theta = \text{diag}(1, -1)$, and $0 < \delta < 2$, then the full Huber multiplier interval is $[-2 + \delta, 2 - \delta]$, while every multiplier produces the same certificate I_2 .

Remark 4.7 (Approximant versus certificate regularization). The canonical trace-class approximation of Legg and Ward [1985] uses the limit of Schatten c_p -approximations as $p \downarrow 1$ to resolve possible nonuniqueness on the approximant side. Our regularization acts on the certificate side under the additional tangency equality. In the no-root regime the approximant is unique by theorem 2.2, whereas the certificate may or may not be unique by theorem 3.4. Thus theorem 4.2 is recovery in the singleton case and genuine Fenchel-center selection otherwise; the rate theory below applies to the corresponding regularized certificate path.

Corollary 4.8 (Global safety bounds). *For every A , (36) holds, and*

$$0 \leq p^* - \langle G, \Phi_\delta \rangle \leq q\delta, \quad |\lambda_\delta| \leq \frac{2\|G\|_* + q\delta}{\|\Theta\|_*}. \quad (47)$$

Proof. For $s \geq 0$, $0 \leq s - (\sqrt{s^2 + \delta^2} - \delta) \leq \delta$; summing proves (36). Since $R \leq q$ on the spectral unit ball, (37) gives the primal bias. Finally, $d_\delta(\lambda_\delta) \leq d_\delta(0) \leq \|G\|_*$, whereas the coercive lower bound used above gives the multiplier estimate. $\square$

The exact interval method needs no smoothing if one merely wants an exact direction. The value of (32) is different: it is a continuous strictly monotone equation, avoids an exact-rank branch, and recovers or selects the explicit variational center. The next section quantifies its behavior at rank loss.

5 Local smoothing laws at regular and singular minimizers

Write

$$A(\lambda) = G + \lambda\Theta, \quad s = \lambda - \lambda^*.$$

All statements in this section are local asymptotics at a fixed instance under the printed nondegeneracy assumptions. The strict rank-deficient law below permits arbitrary shape and corank. Only the boundary and crossover laws use a transverse square corank-one crossing, because their proof isolates one signed Rellich-analytic small mode.

5.1 Full-rank regular minimizers

Theorem 5.1 (Quadratic regular displacement). *Assume that λ^* minimizes ϕ , that $A^* = A(\lambda^*)$ has full rectangular rank, and that $\kappa = \phi''(\lambda^*) > 0$. If $A^* = U\Sigma V^\top$, then*

$$\lambda_\delta - \lambda^* = -\frac{b^*}{\kappa}\delta^2 + o(\delta^2), \quad b^* = -\frac{1}{2}\langle \Theta, U\Sigma^{-2}V^\top \rangle. \quad (48)$$

The matrix in the pairing is basis independent:

$$U\Sigma^{-2}V^\top = \begin{cases} A^*((A^*)^\top A^*)^{-3/2}, & m \geq n, \\ (A^*(A^*)^\top)^{-3/2}A^*, & m < n. \end{cases} \quad (49)$$

Thus the displacement is $O(\delta^2)$, and it is $\Theta(\delta^2)$ when $b^* \neq 0$.

Proof. In a full-rank neighborhood,

$$\frac{\sigma}{\sqrt{\sigma^2 + \delta^2}} = 1 - \frac{\delta^2}{2\sigma^2} + O(\delta^4)$$

uniformly. Functional calculus therefore gives

$$Z_\delta(A(\lambda)) = S_c(A(\lambda)) - \frac{\delta^2}{2} U(\lambda) \Sigma(\lambda)^{-2} V(\lambda)^\top + O(\delta^4).$$

The displayed matrix functions are analytic and basis independent even at repeated positive singular values. Pairing with Θ yields $h_\delta(\lambda) = \phi'(\lambda) + \delta^2 b(\lambda) + O(\delta^4)$, with $b(\lambda^*) = b^*$. The scalar implicit function theorem at $\phi'(\lambda^*) = 0$, $\phi''(\lambda^*) = \kappa$ proves (48). $\square$

5.2 Strict rank-deficient displacement at arbitrary corank

Theorem 5.2 (Strict displacement at arbitrary corank). *Let λ^* minimize ϕ , let $A^* = A(\lambda^*)$ be rank deficient, and define a, C, β by (9). Assume the strict condition $|a| < \beta$. Let $\sigma_1 \geq \dots \geq \sigma_{r_C} > 0$ be the positive singular values of C , and let $\eta_R \geq 0$ be the unique solution of*

$$\sum_{i=1}^{r_C} \frac{\eta_R \sigma_i^2}{\sqrt{1 + \eta_R^2 \sigma_i^2}} = |a|. \quad (50)$$

Then

$$\frac{\lambda_\delta - \lambda^*}{\delta} \longrightarrow -\text{sign}(a)\eta_R. \quad (51)$$

Moreover, with $q = \min\{m, n\}$,

$$|\lambda_\delta - \lambda^*| \leq \frac{q}{\beta - |a|} \delta. \quad (52)$$

No square-shape, corank-one, or rank-one-normal assumption is required. The displacement is $\Theta(\delta)$ exactly when $a \neq 0$; when $a = 0$, it is $o(\delta)$.

Proof. Put $q = \min\{m, n\}$, $r = \text{rank } A^*$, and $t_\delta = (\lambda_\delta - \lambda^*)/\delta$. Strict optimality and theorem 2.2 give the global sharp-growth estimate

$$\phi(\lambda) \geq \phi(\lambda^*) + (\beta - |a|)|\lambda - \lambda^*|. \quad (53)$$

Combining it with (36) and optimality of λ_δ yields (52). Thus (t_δ) is bounded before any local expansion is used.

Define the rescaled convex functions

$$F_\delta(t) = \frac{\Psi_\delta(A(\lambda^* + \delta t)) - \Psi_\delta(A^*)}{\delta}.$$

Choose a spectral gap separating the r positive singular values of A^* from zero. The sum $\ell(s)$ of their analytic separated cluster is smooth even when singular values inside that cluster cross, and $\ell'(0) = \langle \Theta, U_r V_r^\top \rangle = a$. Since $\psi_\delta(\sigma) = \sigma - \delta + O(\delta^2)$ uniformly on the separated cluster, its contribution to $F_\delta(t)$ converges to at , uniformly for t in compact sets.

For the small cluster, let $\Pi_V(s), \Pi_U(s)$ be the Riesz projections of $A(s)^\top A(s)$ and $A(s)A(s)^\top$ onto the clusters converging to zero, where $A(s) = A(\lambda^* + s)$. The fixed gap makes them analytic, with analytic orthonormal frames $V_0(s), U_0(s)$ [Kato, 1995, Chapter II]. Resolvent intertwining gives $A(s)\Pi_V(s) = \Pi_U(s)A(s)$ and its transpose identity, so the cluster subspaces reduce $A(s)$ exactly. Its small singular values are therefore precisely those of

$$B(s) = U_0(s)^\top A(\lambda^* + s) V_0(s) = sC + O(s^2)$$

because $B(0) = 0$ and $B'(0) = U_0^\top \Theta V_0 = C$. Weyl's singular-value bound now gives, uniformly near zero,

$$\sigma_i(A(\lambda^* + s)) = |s| \sigma_i(C) + O(s^2),$$

where zero singular values of C are included. Setting $s = \delta t$ and using that $x \mapsto \sqrt{1+x^2} - 1$ is Lipschitz gives compact-uniform convergence

$$F_\delta(t) \longrightarrow F(t) := at + \sum_{i=1}^{r_C} \left(\sqrt{1 + \sigma_i^2 t^2} - 1 \right). \quad (54)$$

The function F is strictly convex, and $F(t) = at + \beta|t| + o(|t|)$; hence $|a| < \beta$ makes it coercive. Its unique minimizer solves

$$a + \sum_{i=1}^{r_C} \frac{\sigma_i^2 t}{\sqrt{1 + \sigma_i^2 t^2}} = 0,$$

so it equals $-\text{sign}(a)\eta_R$. The localization (52) puts every minimizer of F_δ in one fixed compact interval. Compact-uniform convergence then implies that every accumulation point minimizes F ; uniqueness proves (51). The final order statements follow because $\eta_R > 0$ if and only if $a \neq 0$. $\square$

Remark 5.3 (One scalar governs both path and limit). Throughout this section the subscript R marks objects attached to the pseudo-Huber regularizer, rather than a common tensor type. The multiplier η_R in (51) is exactly the multiplier in the smooth certificate formula (38)–(39). Thus the first displacement of the smoothing path and the kernel block of its limiting certificate are two manifestations of the same scalar balance. More precisely, $K_R = Z_1(t_0 C)$, where $t_0 = -\text{sign}(a)\eta_R$; the next theorem shows that this identity is the zero-scale value of an analytic blow-up path.

Theorem 5.4 (First-order direction law at a strict minimizer). *Under the hypotheses of theorem 5.2, put $t_0 = -\text{sign}(a)\eta_R$. For the fixed instance there is a jointly real-analytic blow-up extension $\mathcal{Z}(\varepsilon, t)$, defined near $(0, t_0)$, such that, for $\varepsilon > 0$,*

$$\mathcal{Z}(\varepsilon, t) = Z_\varepsilon(A(\lambda^\star + \varepsilon t)), \quad \mathcal{Z}(0, t) = U_r V_r^\top + U_0 Z_1(tC) V_0^\top. \quad (55)$$

Let

$$H(\varepsilon, t) = \langle \Theta, \mathcal{Z}(\varepsilon, t) \rangle, \quad d_R := \partial_t H(0, t_0) = \sum_{i=1}^{r_C} \frac{\sigma_i^2}{(1 + t_0^2 \sigma_i^2)^{3/2}} > 0,$$

and set

$$t_1 = -\frac{\partial_\varepsilon H(0, t_0)}{d_R}, \quad \mathcal{M}_R = \partial_\varepsilon \mathcal{Z}(0, t_0) - \frac{\langle \Theta, \partial_\varepsilon \mathcal{Z}(0, t_0) \rangle}{d_R} \partial_t \mathcal{Z}(0, t_0). \quad (56)$$

Then

$$\lambda_\delta - \lambda^\star = \delta t_0 + \delta^2 t_1 + O(\delta^3), \quad (57)$$

$$\Phi_\delta = \Phi_R^\star + \delta \mathcal{M}_R + O(\delta^2). \quad (58)$$

Consequently the returned-direction error is $O(\delta)$, and it is asymptotic to $\delta \|\mathcal{M}_R\|_F$ when $\mathcal{M}_R \neq 0$. If $\mathcal{M}_R = 0$, it is $O(\delta^2)$. In particular, $a = 0$ forces $t_0 = t_1 = 0$ and $\mathcal{M}_R = 0$, so that

$$\lambda_\delta - \lambda^\star = O(\delta^3), \quad \|\Phi_\delta - \Phi_R^\star\|_F = O(\delta^2). \quad (59)$$

The constants are local to the fixed instance; no uniformity as the spectral gap or $\beta - |a|$ tends to zero is asserted.

Proof. Let $r = \text{rank } A^\star$. The gap between its positive and zero singular-value clusters gives local analytic orthogonal frames $U(s) = [U_+(s) \ U_0(s)]$ and $V(s) = [V_+(s) \ V_0(s)]$ for which

$$U(s)^\top A(\lambda^\star + s) V(s) = \begin{pmatrix} A_+(s) & 0 \\ 0 & B(s) \end{pmatrix}.$$

Here $A_+(s) \in \mathbb{R}^{r \times r}$ is invertible, $B(s) \in \mathbb{R}^{(m-r) \times (n-r)}$, and $B(0) = 0$, $B'(0) = C$. Analytic division gives $B(s) = s\tilde{B}(s)$, with $\tilde{B}(0) = C$; unequal left and right nullities and zero singular values of C cause no difficulty. The frames are chosen to agree at $s = 0$ with the fixed compact and null frames used to define a and C .

Write

$$\mathcal{E}_s(K) = U_0(s) K V_0(s)^\top, \quad P_\varepsilon(s) = U_+(s) Z_\varepsilon(A_+(s)) V_+(s)^\top.$$

Orthogonal equivariance, block separation, and the scaling identity $Z_\varepsilon(B) = Z_1(B/\varepsilon)$ give, for $\varepsilon > 0$,

$$Z_\varepsilon(A(\lambda^* + \varepsilon t)) = P_\varepsilon(\varepsilon t) + \mathcal{E}_{\varepsilon t} \left[Z_1 \left(t \tilde{B}(\varepsilon t) \right) \right]. \quad (60)$$

The positive block is analytic in (s, ε^2) , while Z_1 is real analytic on the entire rectangular matrix space. Thus the right-hand side extends jointly analytically across $\varepsilon = 0$ and proves (55). This construction uses analytic cluster subspaces, not analytic individual singular vectors, so repeated positive singular values are allowed [Kato, 1995, Chapter II].

At zero scale,

$$H(0, t) = a + \langle C, Z_1(tC) \rangle = a + \sum_{i=1}^{r_C} \frac{t\sigma_i^2}{\sqrt{1 + t^2\sigma_i^2}}.$$

Its unique zero is t_0 , and differentiation gives the stated positive constant d_R . The analytic implicit function theorem therefore gives $t(\varepsilon) = t_0 + \varepsilon t_1 + O(\varepsilon^2)$. For positive ε , strict convexity and theorem 5.2 identify this local branch with $(\lambda_\varepsilon - \lambda^*)/\varepsilon$. Since $Z_1(t_0 C) = K_R$, Taylor expansion of $\mathcal{Z}(\delta, t(\delta))$ proves (57)–(58) and (56). When $a = 0$, one has $t_0 = 0$, while $Z_\varepsilon(A^*) = S_c(A^*) + O(\varepsilon^2)$. Hence $\partial_\varepsilon H(0, 0) = 0$ and $\partial_\varepsilon \mathcal{Z}(0, 0) = 0$, proving (59). $\square$

Proposition 5.5 (Intrinsic computable first-correction formula). *Use the analytic cluster frames in the proof of theorem 5.4, write $\tilde{B}(s) = B(s)/s$ with its analytic value $\tilde{B}(0) = C$, and define the basis-free frozen-scale field*

$$\mathcal{W}(s, t) = P_{\text{big}}(s) + U_0(s)Z_1(t\tilde{B}(s))V_0(s)^\top, \quad P_{\text{big}}(s) = U_+(s)S_c(A_+(s))V_+(s)^\top. \quad (61)$$

Then $\mathcal{W}(0, t_0) = \Phi_R^*$ and

$$\partial_\varepsilon \mathcal{Z}(0, t_0) = t_0 \partial_s \mathcal{W}(0, t_0). \quad (62)$$

Consequently the constants in (56) are

$$t_1 = -\frac{t_0 \langle \Theta, \partial_s \mathcal{W}(0, t_0) \rangle}{d_R}, \quad (63)$$

$$\mathcal{M}_R = t_0 \left[\partial_s \mathcal{W}(0, t_0) - \frac{\langle \Theta, \partial_s \mathcal{W}(0, t_0) \rangle}{d_R} \partial_t \mathcal{W}(0, t_0) \right]. \quad (64)$$

In particular, $t_0 = 0$ makes both constants vanish. If $t_0 \neq 0$, then

$$\begin{aligned} \mathcal{M}_R = 0 &\iff \partial_s \mathcal{W}(0, t_0) \in \text{span}\{\partial_t \mathcal{W}(0, t_0)\}, \\ t_1 = 0 &\iff \langle \Theta, \partial_s \mathcal{W}(0, t_0) \rangle = 0, \end{aligned}$$

and both vanish if and only if $\partial_s \mathcal{W}(0, t_0) = 0$.

Proof. Both terms in (61) are invariant under analytic orthogonal changes of the left and right cluster frames, by orthogonal equivariance of Z_1 . In (60), the positive block satisfies $P_\varepsilon(s) = P_{\text{big}}(s) + O(\varepsilon^2)$, jointly analytically in (s, ε^2) . All remaining ε -dependence enters through $s = \varepsilon t$. Hence

$$\mathcal{Z}(\varepsilon, t) = \mathcal{W}(\varepsilon t, t) + O(\varepsilon^2),$$

and the chain rule proves (62). Substitution into (56) gives (63)–(64). Finally, $\langle \Theta, \partial_t \mathcal{W}(0, t_0) \rangle = d_R > 0$; the stated vanishing alternatives follow directly. Notice in particular that parallelism alone cancels $\mathcal{M}_R$, but need not cancel t_1 . The full-rank polar-block derivatives in this formula may be evaluated by standard Fréchet-derivative algorithms [Gawlik and Leok, 2017]; the new feature here is their coupling to the analytic rank-deficient blow-up. $\square$

Corollary 5.6 (Sharp compact-root superconvergence). *Under theorem 5.4, suppose $a = 0$. By theorem 2.2, this is exactly compact-polar stationarity at λ^* . Let*

$$J_\star = U_r \Sigma_r^{-2} V_r^\top, \quad d_0 = \|C\|_F^2, \quad \chi_\star = \langle \Theta, J_\star \rangle. \quad (65)$$

where $J_\star = 0$ when the positive singular block is empty. Strictness gives $C \neq 0$, hence $d_0 > 0$. Then $\Phi_R^* = S_c(A^*)$, and

$$\lambda_\delta - \lambda^* = \frac{\chi_\star}{2d_0} \delta^3 + O(\delta^4), \quad (66)$$

$$\Phi_\delta - \Phi_R^* = \delta^2 \mathcal{N}_0 + O(\delta^3), \quad \mathcal{N}_0 = -\frac{1}{2} J_\star + \frac{\chi_\star}{2d_0} U_0 C V_0^\top. \quad (67)$$

Moreover,

$$\|\mathcal{N}_0\|_F^2 = \frac{1}{4}\|J_\star\|_F^2 + \frac{\chi_\star^2}{4d_0}. \quad (68)$$

Thus the multiplier error is exactly cubic when $\chi_\star \neq 0$, and the direction error is exactly quadratic whenever $\text{rank } A^\star > 0$. If $\chi_\star = 0$, put

$$J_2 = U_r \Sigma_r^{-4} V_r^\top, \quad \chi_2 = \langle \Theta, J_2 \rangle. \quad (69)$$

Then the next coefficients are

$$\lambda_\delta - \lambda^\star = -\frac{3\chi_2}{8d_0} \delta^5 + O(\delta^6), \quad (70)$$

$$\Phi_\delta - \Phi_R^\star = -\frac{\delta^2}{2} J_\star + \delta^4 \left[\frac{3}{8} J_2 - \frac{3\chi_2}{8d_0} U_0 C V_0^\top \right] + O(\delta^5). \quad (71)$$

In particular, the multiplier error is exactly quintic if $\chi_2 \neq 0$.

Proof. Here $t_0 = 0$, $d_R = d_0 > 0$, and

$$\mathcal{Z}(\varepsilon, 0) = \mathcal{Z}_\varepsilon(A^\star) = S_c(A^\star) - \frac{\varepsilon^2}{2} J_\star + O(\varepsilon^4), \quad \partial_t \mathcal{Z}(0, 0) = U_0 C V_0^\top.$$

It follows that $H(\varepsilon, 0) = -\chi_\star \varepsilon^2/2 + O(\varepsilon^4)$. The analytic implicit solution therefore satisfies

$$t(\varepsilon) = \frac{\chi_\star}{2d_0} \varepsilon^2 + O(\varepsilon^3).$$

Since $\lambda_\varepsilon - \lambda^\star = \varepsilon t(\varepsilon)$, substitution into the analytic blow-up gives (66)–(67). The range–range and null–null blocks in $\mathcal{N}_0$ are Frobenius orthogonal, which proves (68). If $\chi_\star = 0$, the separated positive block has the uniform expansion

$$\frac{\sigma}{\sqrt{\sigma^2 + \varepsilon^2}} = 1 - \frac{\varepsilon^2}{2\sigma^2} + \frac{3\varepsilon^4}{8\sigma^4} + O(\varepsilon^6).$$

Consequently $H(\varepsilon, 0) = 3\chi_2 \varepsilon^4/8 + O(\varepsilon^6)$, and the implicit branch satisfies $t(\varepsilon) = -3\chi_2 \varepsilon^4/(8d_0) + O(\varepsilon^5)$. This proves (70). Now $s = \varepsilon t(\varepsilon) = O(\varepsilon^5)$. The quartic term of the positive block and $t(\varepsilon) \partial_t \mathcal{Z}(0, 0)$, with $\partial_t \mathcal{Z}(0, 0) = U_0 C V_0^\top$, give (71). Indeed, analyticity yields $\mathcal{Z}(\varepsilon, t(\varepsilon)) = \mathcal{Z}(\varepsilon, 0) + t(\varepsilon) \partial_t \mathcal{Z}(0, 0) + O(\varepsilon |t(\varepsilon)| + t(\varepsilon)^2)$, so every remaining term is $O(\varepsilon^5)$. $\square$

Remark 5.7 (An odd-order multiplier hierarchy). The cubic and quintic laws are the first two levels of a complete multiplier hierarchy. For an integer $k \geq 1$, let

$$b_k = (-1)^k 4^{-k} \binom{2k}{k}, \quad J_k = U_r \Sigma_r^{-2k} V_r^\top, \quad \chi_k = \langle \Theta, J_k \rangle.$$

If an integer $p \geq 1$ satisfies $a = 0$ and $\chi_1 = \dots = \chi_{p-1} = 0$, then the binomial expansion used above and the analytic implicit function theorem give

$$\lambda_\delta - \lambda^\star = -\frac{b_p \chi_p}{d_0} \delta^{2p+1} + O(\delta^{2p+2}). \quad (72)$$

Thus each further orthogonality cancellation raises the multiplier by two orders; $p = 1$ is (66) and $p = 2$ is (70). This hierarchy is local to the fixed strict crossing and does not assert that the cancellations are generic.

At a fixed strict rank-deficient minimizer, these results give a sharp dichotomy. Compact-polar stationarity holds exactly when $a = 0$, in which case smoothing has cubic multiplier bias and quadratic direction bias with the refinements above. If $a \neq 0$, compact-polar stationarity fails and the multiplier bias is necessarily $\Theta(\delta)$; the direction bias is $\Theta(\delta)$ when $\mathcal{M}_R \neq 0$, but may cancel to second order.

5.3 A transverse corank-one normal form for boundary and crossover

For the remaining results assume $m = n = q$, $\text{rank } A^* = q - 1$, and let u_0, v_0 be unit left and right null vectors. Put

$$\alpha = u_0^\top \Theta v_0, \quad \beta = |\alpha|, \quad \alpha \neq 0. \quad (73)$$

Lemma 5.8 (Analytic kink normal form). *There are real-analytic functions τ, ℓ near zero such that*

$$\tau(s) = \alpha s + O(s^2), \quad \phi(\lambda^* + s) = \ell(s) + |\tau(s)|. \quad (74)$$

Moreover,

$$h_\delta(\lambda^* + s) = \ell'_\delta(s) + \frac{\tau(s)\tau'(s)}{\sqrt{\tau(s)^2 + \delta^2}}, \quad \ell'_\delta(s) = \ell'(s) + O(\delta^2) \quad (75)$$

uniformly on a fixed small neighborhood, where $\ell'_\delta(s)$ denotes the contribution to h_δ from the separated nonzero singular-value cluster. With $a = \ell'(0)$, the exact subdifferential is $[a - \beta, a + \beta]$.

Proof. Consider the analytic symmetric dilation

$$H(s) = \begin{pmatrix} 0 & A^* + s\Theta \\ (A^* + s\Theta)^\top & 0 \end{pmatrix}.$$

Its zero eigenspace has basis $(u_0, 0), (0, v_0)$, and the compression of $H'(0)$ is $\begin{pmatrix} 0 & \alpha \\ \alpha & 0 \end{pmatrix}$. Rellich perturbation theory [Kato, 1995, Chapter II] therefore gives a signed analytic pair $\pm\tau(s)$ with $\tau'(0) = \alpha$. The small singular value is $|\tau(s)|$. The sum of the separated positive eigenvalue cluster of the dilation is analytic and defines $\ell(s)$, proving (74). The nonzero cluster stays uniformly separated from zero, so the regular expansion used in theorem 5.1 gives $\ell'_\delta = \ell' + O(\delta^2)$. Differentiating the smoothed small mode proves (75); the two one-sided derivatives at zero are $a \mp \beta$. $\square$

5.4 The corank-one strict specialization and boundary kink

Corollary 5.9 (Linear strict-kink displacement, corank one). *Under lemma 5.8, suppose $|a| < \beta$. Then*

$$\frac{\lambda_\delta - \lambda^*}{\delta} \rightarrow t_0 = -\frac{a}{\beta\sqrt{\beta^2 - a^2}}. \quad (76)$$

The displacement is $O(\delta)$, and it is $\Theta(\delta)$ when $a \neq 0$.

Proof. Here C has the single positive singular value β . Equation (50) gives $\eta_R = |a|/[\beta\sqrt{\beta^2 - a^2}]$, so theorem 5.2 reduces to (76). $\square$

At the boundary, the small-mode expansion must be grouped with the complete contact-side derivative; expanding the mode separately loses a linear term. Here ϕ'_{con} denotes the derivative of the full-rank analytic branch continued from the stated contact side. The boundary law is not obtained by sending $|a| \uparrow \beta$ in the strict formula: $\eta_R \rightarrow \infty$, so the δ -scale balance itself breaks down and is replaced by the $\delta^{2/3}$ scale below.

Theorem 5.10 (Boundary-kink two-thirds law). *Suppose $|a| = \beta$, and choose the sign of the analytic branch in lemma 5.8 so that $\alpha = \beta > 0$. Let the full-rank contact side be $s < 0$ when $a = \beta$ and $s > 0$ when $a = -\beta$, and let ϕ_{con} denote the analytic branch continued from that side. Then lemma 5.8 gives*

$$\phi'_{\text{con}}(\lambda^* + s) = \kappa s + O(s^2), \quad \kappa := \begin{cases} \ell''(0) - \tau''(0), & a = \beta, \\ \ell''(0) + \tau''(0), & a = -\beta. \end{cases} \quad (77)$$

Convexity gives $\kappa \geq 0$; assume only the nondegeneracy $\kappa > 0$. Then

$$\lambda_\delta - \lambda^* = -\text{sign}(a) \left(\frac{1}{2\beta\kappa} \right)^{1/3} \delta^{2/3} (1 + o(1)). \quad (78)$$

Proof. The expansion in (77) is a consequence, not an extra hypothesis. On the contact side one has $\phi_{\text{con}} = \ell - \tau$ when $a = \beta$ and $\phi_{\text{con}} = \ell + \tau$ when $a = -\beta$. Hence its derivative at zero is respectively $a - \beta = 0$ or $a + \beta = 0$, and analyticity gives the displayed expansion with the stated second derivative.

It suffices to treat $a = \beta$; the other case follows after reversing s and the signed analytic branch. Since $\tau(0) = 0$, (75) gives $h_\delta(\lambda^*) = \ell'_\delta(0) = a + O(\delta^2) > 0$ for all small δ , and h_δ is strictly increasing; hence the smooth root lies on $s < 0$. Put $\chi(s) = \text{sign}(\tau(s))\tau'(s)$. Then $\chi(s) = -\beta + O(s)$ and $\tau(s)^2 = \beta^2 s^2(1 + O(s))$ on that side. In the regime $\delta/|s| \rightarrow 0$, group the exact contact derivative before expanding:

$$h_\delta(\lambda^* + s) = \phi'_{\text{con}}(\lambda^* + s) + \frac{\delta^2}{2\beta s^2}(1 + O(|s|)) \quad (79)$$

$$+ O(\delta^2) + O(\delta^4/s^4). \quad (80)$$

Set $s = -c\delta^{2/3}$ and divide by $\delta^{2/3}$. Every remainder in (80) vanishes, and the limiting equation is

$$-\kappa c + \frac{1}{2\beta c^2} = 0.$$

Its positive root is $c_0 = (2\beta\kappa)^{-1/3}$. Evaluating at fixed $c_- < c_0 < c_+$ gives opposite signs for all small δ , so scalar monotonicity brackets the unique root. This also proves $|s_\delta| \asymp \delta^{2/3} \gg \delta$, validating the expansion. Letting the brackets approach c_0 proves (78). $\square$

The curvature condition is genuine. For $A(\lambda^* + s) = \text{diag}(1 + s, s)$, one has $a = \beta = 1$, but the contact branch is constant on $-1 < s < 0$, so $\kappa = 0$ and there is no isolated two-thirds balance.

For the orientation $a = \beta > 0$ used in the crossover theorem below, this contact curvature is exactly $\kappa = \ell''(0) - \tau''(0) = \kappa_0$.

For the boundary member of theorem 3.1 with $b_1 = b_2 = c = 1$ and $d = 0$, one has $\lambda^* = 1$, $a = \beta = 1/2$, and $\phi'_{\text{con}}(1 + s) = s/\sqrt{s^2 + 4} = s/2 + O(s^3)$ on the left. Hence

$$1 - \lambda_\delta \sim 2^{1/3}\delta^{2/3}. \quad (81)$$

The exponent is not intrinsic to the certificate slice alone: it also records how the chosen profile approaches the spectral-ball boundary.

Corollary 5.11 (The boundary exponent is a tail law). *Under the hypotheses of theorem 5.10, let λ_δ^ψ be any minimizer of the objective induced by an admissible profile from proposition 4.5.*

1. If, for some $p > 1$ and $\gamma_\psi > 0$,

$$\psi'(x) = 1 - \gamma_\psi x^{-p} + o(x^{-p}) \quad (x \rightarrow \infty), \quad (82)$$

then

$$\lambda_\delta^\psi - \lambda^* = -\text{sign}(a) \left(\frac{\gamma_\psi}{\kappa\beta^{p-1}} \right)^{1/(p+1)} \delta^{p/(p+1)} (1 + o(1)). \quad (83)$$

2. If ψ' first reaches one at a finite threshold $L_\psi > 0$ and is identically one thereafter, then

$$\lambda_\delta^\psi - \lambda^* = -\text{sign}(a) \frac{L_\psi}{\beta} \delta + o(\delta). \quad (84)$$

For the standard Huber profile in (45), $L_\psi = 1$ and the sharper expansion is

$$\lambda_\delta^{\text{hub}} - \lambda^* = -\text{sign}(a) \frac{\delta}{\beta} + O(\delta^2). \quad (85)$$

Pseudo-Huber has $p = 2$ and $\gamma_\psi = 1/2$, so (83) recovers (78); finite saturation yields the separate linear boundary regime.

Proof. It is enough to take $a = \beta > 0$, for which the contact side is $s < 0$. Along a scale with $\delta/|s| \rightarrow 0$, the separated positive cluster differs from its polar limit by $o(|s|)$ on the scale obtained below. Using $\tau(s) = \beta s + O(s^2)$ and grouping against the exact contact derivative, (82) gives

$$h_\delta^\psi(\lambda^* + s) = \kappa s + \frac{\gamma_\psi \delta^p}{\beta^{p-1}|s|^p} + o\left(|s| + \frac{\delta^p}{|s|^p}\right).$$

Putting $s = -c\delta^{p/(p+1)}$ and bracketing the monotone root yields $\kappa c^{p+1} = \gamma_\psi/\beta^{p-1}$, proving the first claim.

If saturation first occurs at L_ψ , the separated modes are already saturated for all small δ . Monotonicity brackets the root immediately inside the transition edge $|\tau(s)| = L_\psi\delta$, and therefore $|s|/(L_\psi\delta/\beta) \rightarrow 1$, proving (84). For standard Huber, inside $|\tau(s)| < \delta$ the small mode gives the sharper expansion

$$h_\delta^{\text{hub}}(\lambda^* + s) = \beta + \frac{\beta^2 s}{\delta} + O(|s| + s^2/\delta).$$

The monotone root therefore satisfies $s = -\delta/\beta + O(\delta^2)$. Reflection handles $a = -\beta$. $\square$

Corollary 5.12 (Selected-direction upper bounds). *In fixed dimension and under the corresponding hypotheses of theorems 5.1, 5.4 and 5.10,*

$$\|\Phi_\delta - \Phi_R^*\|_F = O(\delta^2), \quad O(\delta), \quad O(\delta^{2/3}), \quad (86)$$

respectively. These are upper bounds; exact matrix-direction rates require a corresponding noncancellation condition. The strict entry holds for arbitrary rectangular shape and corank; theorem 5.4 gives its first-order constant and the sharper $O(\delta^2)$ rate when $a = 0$.

Proof. On the separated cluster, the polar factor and smoothed spectral map are analytic in the matrix argument and δ^2 , so their variation is the multiplier displacement plus $O(\delta^2)$; this proves the regular entry. The strict entry is (58). At a boundary kink, both the singular-vector movement and the coefficient error are $O(\delta^{2/3})$ because $|s| = O(\delta^{2/3})$ and $\delta^2/s^2 = O(\delta^{2/3})$. $\square$

5.5 A local strict-to-boundary crossover

The fixed-instance constants in theorem 5.4 and proposition 5.5 deteriorate as $\beta - |a| \downarrow 0$: η_R diverges and d_R vanishes. The result below is the uniform replacement in precisely that degenerating direction, currently for a transverse square corank-one crossing. Thus the boundary law is not confined to one isolated parameter value; a moving normal form controls the entire nearby transition.

Lemma 5.13 (Moving crossing for general matrix normals). *Let $A(\lambda, z) = G(z) + \lambda\Theta(z)$, $z \in \mathbb{R}^p$, where G and Θ are real analytic and $\Theta(z) \neq 0$. Suppose the matrices are square, $A(\lambda_0, 0)$ has corank one, and*

$$u_0^\top \Theta(0)v_0 \neq 0. \quad (87)$$

After shrinking the neighborhood, there are an analytic crossing $\lambda_{\text{sing}}(z)$ and jointly analytic functions τ, ℓ, w , with $s = \lambda - \lambda_{\text{sing}}(z)$, such that

$$\|A(\lambda_{\text{sing}}(z) + s, z)\|_* = \ell(s, z) + |\tau(s, z)|, \quad \tau(s, z) = s\sqrt{w(s, z)}, \quad w(s, z) > 0. \quad (88)$$

The sign can be chosen so that $\beta(z) = \partial_s \tau(0, z) > 0$, and ℓ is analytic even if singular values inside the separated positive cluster cross.

Proof. At a corank-one matrix the adjugate is a nonzero multiple of $v_0 u_0^\top$. Condition (87) makes $\partial_\lambda \det A(\lambda_0, 0) \neq 0$, so the analytic implicit function theorem gives $\lambda_{\text{sing}}(z)$. Set $\tilde{A}(s, z) = A(\lambda_{\text{sing}}(z) + s, z)$ and $B(s, z) = \tilde{A}(s, z)^\top \tilde{A}(s, z)$. For each nearby z , $B(0, z)$ has a simple zero eigenvalue separated from the rest; its analytic continuation $\mu(s, z)$ satisfies $\mu(0, z) = 0$. Analytic division gives $\mu = s\mu_1$. Since $B \succeq 0$, the scalar function $s \mapsto \mu(s, z)$ has a minimum at zero, so a second division gives $\mu = s^2 w$. Transversality makes $w > 0$ after shrinking.

For the remaining modes, a fixed contour around the positive cluster of the symmetric dilation defines a jointly analytic Riesz projection. The trace of the dilation against this projection is the analytic eigenvalue sum $\ell(s, z)$, proving (88). $\square$

Theorem 5.14 (Unified cubic-root crossover). *In lemma 5.13, suppose at $z = 0$*

$$a_0 = \partial_s \ell(0, 0) = \beta_0 > 0, \quad \kappa_0 = \partial_{ss} \ell(0, 0) - \partial_{ss} \tau(0, 0) > 0. \quad (89)$$

Let $z = z_\delta \rightarrow 0$ approach from the strict or boundary side and set

$$\begin{aligned} a_\delta &= \partial_s \ell(0, z_\delta), & \beta_\delta &= \partial_s \tau(0, z_\delta), \\ \Delta_\delta &= \beta_\delta - a_\delta \geq 0, & \kappa_\delta &= \partial_{ss} \ell(0, z_\delta) - \partial_{ss} \tau(0, z_\delta). \end{aligned}$$

Let $\lambda_\delta^* = \lambda_{\text{sing}}(z_\delta)$, let λ_δ be the smoothed minimizer, and put $x_\delta = \lambda_\delta^* - \lambda_\delta$. Define $\hat{x}_\delta > 0$ as the unique positive root of

$$\kappa_\delta \hat{x}_\delta^3 + \Delta_\delta \hat{x}_\delta^2 = \frac{\delta^2}{2\beta_\delta}. \quad (90)$$

Then $x_\delta > 0$ for all small δ , and, along every such path, including oscillatory ones,

$$\frac{x_\delta}{\hat{x}_\delta} = 1 + O(\Delta_\delta + \hat{x}_\delta). \quad (91)$$

Consequently:

1. if $\Delta_\delta/\delta^{2/3} \rightarrow \zeta \in [0, \infty)$, then

$$\frac{x_\delta}{\delta^{2/3}} \rightarrow y_\zeta, \quad \kappa_0 y_\zeta^3 + \zeta y_\zeta^2 = \frac{1}{2\beta_0}; \quad (92)$$

2. if $\Delta_\delta/\delta^{2/3} \rightarrow \infty$, then

$$\frac{x_\delta \sqrt{\Delta_\delta}}{\delta} \rightarrow \frac{1}{\sqrt{2\beta_0}}; \quad (93)$$

3. for arbitrary paths,

$$x_\delta \asymp \hat{x}_\delta \asymp \min \left\{ \frac{\delta}{\sqrt{\Delta_\delta}}, \delta^{2/3} \right\}, \quad (94)$$

where the first entry is $+\infty$ at $\Delta_\delta = 0$.

Proof. The one-sided derivatives at the moving crossing are $-\Delta_\delta \leq 0$ and $a_\delta + \beta_\delta \rightarrow 2\beta_0 > 0$, so λ_δ^* is an exact minimizer. For the instance indexed by z , set

$$h_{\delta,z}(\lambda) = \langle \Theta(z), Z_\delta(A(\lambda, z)) \rangle, \quad \Delta(z) = \beta(z) - a(z), \quad \kappa(z) = \partial_{ss} \ell(0, z) - \partial_{ss} \tau(0, z),$$

and $H_{\delta,z}(x) = h_{\delta,z}(\lambda_{\text{sing}}(z) - x)$. Joint analyticity and the grouped expansion in (80) give constants $z_0, x_0, \eta_0, C > 0$, independent of the path, such that whenever $\|z\| \leq z_0$, $0 < x \leq x_0$, and $0 < \delta \leq \eta_0 x$,

$$H_{\delta,z}(x) = -\Delta(z) - \kappa(z)x + \frac{\delta^2}{2\beta(z)x^2} + E_{\delta,z}(x), \quad (95)$$

$$|E_{\delta,z}(x)| \leq C \left(x^2 + \frac{\delta^2}{x} + \delta^2 + \frac{\delta^4}{x^4} \right). \quad (96)$$

At $x = 0$, $H_{\delta,z}(0) = a_\delta + O(\delta^2) > 0$, whereas it is negative at a sufficiently small fixed $x_0 > 0$. Strict convexity places the unique root in $(0, x_0)$.

Put $M_\delta = \Delta_\delta + \kappa_\delta \hat{x}_\delta$. Equation (90) is equivalent to

$$\frac{\delta^2}{2\beta_\delta \hat{x}_\delta^2} = M_\delta.$$

It implies $\hat{x}_\delta, M_\delta \rightarrow 0$ and $\delta/\hat{x}_\delta = \sqrt{2\beta_\delta M_\delta} \rightarrow 0$, so the uniform expansion applies at $x = c\hat{x}_\delta$. Uniformly for c in a compact subset of $(0, \infty)$, the four terms in the remainder, divided by M_δ , are respectively

$$O(\hat{x}_\delta), \quad O(\hat{x}_\delta), \quad O(\hat{x}_\delta^2), \quad O(M_\delta).$$

The leading part at this point is

$$\Delta_\delta(c^{-2} - 1) + \kappa_\delta \hat{x}_\delta(c^{-2} - c).$$

It is a fixed positive multiple of M_δ for fixed $c < 1$, and a fixed negative multiple for fixed $c > 1$. Monotonicity therefore gives $x_\delta/\hat{x}_\delta \rightarrow 1$. More quantitatively, for $c = 1 + u$ with $|u|$ small, the leading term has the sign of $-u$ and magnitude at least $c_0|u|M_\delta$, whereas the remainder is bounded by $C_1(\Delta_\delta + \hat{x}_\delta)M_\delta$, with uniform positive constants c_0, C_1 . Taking $u = \pm K(\Delta_\delta + \hat{x}_\delta)$ and $K > C_1/c_0$ gives the quantitative bracket (91).

The three consequences now follow directly from (90): divide it by δ^2 in the critical window, drop its cubic term when $\Delta_\delta/\delta^{2/3} \rightarrow \infty$, and compare the two nonnegative terms for the two-sided order bound. $\square$

The cubic also controls the matrix actually returned by the oracle. The comparator must be the center of the moving instance rather than the fixed center at $z = 0$.

Corollary 5.15 (Direction-level crossover). *Under theorem 5.14, let $P(s, z)$ be the analytic polar contribution of the separated positive singular cluster and let $D(s, z)$ be the analytic signed dyad of the small mode, chosen so that $\langle \Theta(z), D(0, z) \rangle = \beta(z) > 0$. With derivatives evaluated at $(s, z) = (0, 0)$, define*

$$\mathcal{M}_0 = -\partial_s P(0, 0) + \partial_s D(0, 0) + \frac{\kappa_0}{\beta_0} D(0, 0). \quad (97)$$

Then, relative to the exact Fenchel center $\Phi_R^*(z_\delta)$ of the moving instance,

$$\frac{\Phi_\delta - \Phi_R^*(z_\delta)}{x_\delta} \longrightarrow \mathcal{M}_0. \quad (98)$$

Hence

$$\|\Phi_\delta - \Phi_R^*(z_\delta)\|_F = O(\widehat{x}_\delta),$$

and if $\mathcal{M}_0 \neq 0$, this norm is asymptotic to $\|\mathcal{M}_0\|_F \widehat{x}_\delta$; if $\mathcal{M}_0 = 0$, it is $o(\widehat{x}_\delta)$.

Proof. The smoothed spectral decomposition at $s = -x_\delta$ is

$$\Phi_\delta = P_\delta(-x_\delta, z_\delta) + r_\delta(-x_\delta, z_\delta) D(-x_\delta, z_\delta), \quad r_\delta = \frac{\tau}{\sqrt{\tau^2 + \delta^2}},$$

where $P_\delta = P + O(\delta^2)$. Tangency gives $r_\delta = -\ell'_\delta/\tau'$. Expanding this quotient and using $a = \beta - \Delta$ and $\kappa = \ell_{ss} - \tau_{ss}$ gives, uniformly,

$$r_\delta(-x, z) + \frac{a(z)}{\beta(z)} = x \left\{ \frac{\kappa(z)}{\beta(z)} + \frac{\Delta(z)\tau_{ss}(0, z)}{\beta(z)^2} \right\} + O(x^2 + \delta^2).$$

Since $\Delta(z_\delta) \rightarrow 0$ and $\delta^2/x_\delta \rightarrow 0$, this yields

$$\frac{r_\delta(-x_\delta, z_\delta) + a(z_\delta)/\beta(z_\delta)}{x_\delta} \longrightarrow \frac{\kappa_0}{\beta_0}.$$

Since $\Phi_R^*(z) = P(0, z) - [a(z)/\beta(z)]D(0, z)$, Taylor expansion proves (98); the separated-cluster smoothing error is negligible because $\delta^2/x_\delta \rightarrow 0$. The norm statements follow from (91). $\square$

The moving-crossing recentering is essential: nonnegativity together with a single identity $\mu(0, 0) = 0$ would not imply the factorization $\mu = s^2 w$. The theorem is uniform only near one fixed analytic transverse corank-one boundary instance.

6 Numerical verification of the constants

The local laws predict constants, not only exponents. Assertion-based scripts therefore test closed-form instances at high precision; the full tables and execution details are in the technical supplement. For the fixed 2×2 family $G = \begin{pmatrix} 0 & b_1 \\ b_2 & c \end{pmatrix}$, $\Theta = e_1 e_1^\top$, the three basic normalized limits are:

Regime	predicted	computed at the smallest δ
regular	$-16/9 = -1.777777778$	-1.777777777
strict corank one	$-5/(4\sqrt{15}) = -0.3227486122$	-0.3227486102
boundary	$2^{1/3} = 1.259921050$	1.259921042

On the same boundary instance, standard Huber instead gives $(1 - \lambda_\delta^{\text{hub}})/\delta = 1.999999997$ at $\delta = 10^{-8}$, versus the prediction $1/\beta = 2$ in (85). For the admissible profile $\psi'(x) = x/(1 + x^3)^{1/3}$ on $x \geq 0$, the predicted $p = 3$ constant is 1.2778862085; the computed normalized displacement is 1.2778860473 at $\delta = 10^{-9}$. This checks that the exponent changes with the tail, not merely between the two named profiles.

For a corank-two check, take $A^* = \text{diag}(1, 1, 0, 0)$, $\Theta = \text{diag}(3/8, 3/8, 1, 1/2)$, and $\lambda^* = 1$. Here $\eta_R = 0.715080068980$ and $\|K_F - K_R\|_F = 0.040996138377$. At $\delta = 10^{-4}$, the pseudo-Huber slope is -0.7150801 , while Huber gives the water-filling slope -0.6000000000 and recovers Φ_F^* to 10^{-11} . Because $P_{\text{big}}(s)$ and $\tilde{B}(s)$ are constant on this diagonal line, $\partial_s \mathcal{W} = 0$ proves $t_1 = \mathcal{M}_R = 0$. A non-diagonal rational instance instead predicts $\|\mathcal{M}_R\|_F = 0.671775381$, versus the scaled value 0.671776068 at $\delta = 10^{-6}$.

The compact-root instance $A^* = \text{diag}(2, 1, 0)$, $\Theta = \text{diag}(1, -1, 1)$ verifies the predicted cubic coefficient $-3/8$ and quadratic direction norm $\sqrt{26}/8$. After the second cancellation, the rational 4×4 instance in the supplement has $\chi_2 = 1/3$ and converges monotonically to the quintic coefficient $-1/8$; its normalized quartic direction converges to the full matrix in (71).

The moving-family assertions cover boundary, critical, strict-side, and oscillatory paths and verify both the cubic root and direction constants. Complete data, matrices, and finite-scale sequences are in the supplement; the checks test constants but imply no occurrence-frequency law.

7 Discussion and limitations

The two remedies make different tradeoffs. Exact interval bisection followed by (22) has no smoothing bias but needs a rank decision. Pseudo-Huber gives a strictly monotone equation and converges to K_R , whereas the family in proposition 4.5 identifies K_F as the Huber Fenchel center. These centers agree only in the cases of corollary 4.4. The boundary bias is profile dependent: polynomial tail order p gives $\delta^{p/(p+1)}$, pseudo-Huber gives $\delta^{2/3}$, and finite saturation gives δ . At fixed strict crossings the first direction coefficient is computable from (61); compact-root instances instead begin at cubic/quadratic order and obey the multiplier hierarchy of remark 5.7.

The strict laws allow arbitrary shape and corank, but boundary and crossover laws currently require a transverse square corank-one crossing; multiple signed small modes and nontransverse exponents remain open. The admissible family does not cover every vanishing regularizer, and full-space openness implies no frequency law without a distribution for (G, Θ) . No outer complexity or learning-performance gain follows without a separate descent model and control of the changing normal.

8 Conclusion

We characterized the tangency-constrained certificate slice for nuclear-norm approximation from a matrix line and identified its hard and Fenchel spectral centers. An admissible smoothing family links center selection and boundary exponents, while strict arbitrary-corank paths admit computable analytic multiplier and direction expansions. Transverse corank-one boundaries exhibit the sharp pseudo-Huber two-thirds law and a uniform cubic crossover. Compact-polar open failure explains why the full certificate slice is essential.

Data and code availability

No external data are used. Assertion-based Python scripts reproducing every numerical table and constant are included in the supplementary source archive submitted with the manuscript; the archive also records the required Python, NumPy, SciPy, and mpmath dependencies.

Declaration on generative AI assistance

Generative AI tools assisted with language editing, literature discovery, algebraic exploration, drafting portions of theorem and proof text, and drafting verification code. The authors independently checked every cited source, proof, and computational result and assume responsibility for all content.

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